The thermodiffusive lens in lean premixed hydrogen flames

Measurements, derivations and the analysis procedure

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Abstract

Cellular structures in lean premixed hydrogen flames are characterised here through the joint statistics of temperature and fuel mass fraction, taken in range space rather than on any particular definition of a "cell". The principal findings are: the lens structures visible between crossing T and Y_{H_2} contours are the *hot-and-rich* quadrant of the T–Y_{H_2} plane, and they capture fuel consumption to within 7% of the best region of equal volume that could be drawn; they are a level set of the excess-enthalpy scalar $S = \theta + y - 1$, so a single threshold replaces the two; the Lewis number measured from local flux ratios is $\mathrm{Le} = 0.313$; and at fixed temperature the fuel mass fraction varies by a factor of 2.2 with flame curvature, with consumption following it. The curvature, lens and band-shape observations are three views of one differential-diffusion imbalance. All numbers come from a single plotfile of a four-level AMR simulation at one instant; the procedure for reproducing and extending them is given in Sections 8–9.

Contents

1 What was measured, and on what **2**

2 Coarse-fine-aware gradients **2**
2.1 Verification . . . 3

3 The lens **3**
3.1 Orientation . . . 3
3.2 Monotonicity . . . 3
3.3 Efficiency against the best possible region . . . 3
3.4 Optimality condition . . . 4
3.5 A one-parameter family . . . 4
3.6 What the angle of intersection does not add . . . 4

4 The differential-diffusion postulate **5**
4.1 Statement . . . 5
4.2 Test and result . . . 5
4.3 The gradient scaling: supported, with a gap . . . 5

5 The Lewis number **6**
5.1 Two estimators from the same first integrals . . . 6
5.2 They disagree, and why . . . 6
5.3 Apparent Lewis number as a diagnostic . . . 6

5.4 Temperature dependence 7

6 Curvature and resolution **7**

7 Synthesis **8**

8 Procedure **8**
8.1 Build 8
8.2 Verify 9
8.3 Run 9
8.4 Analyse an existing table 9

9 Porting to a larger machine **9**
9.1 Resources 9
9.2 GPU status, honestly 10
9.3 Transfer checklist 10

10 Open items **11**

1 What was measured, and on what

The data is a single plotfile, `plt29791`, from a lean premixed hydrogen flame: a $1.86 \times 1.86 \times 3.72$ mm domain, periodic in x and y, four AMR levels with a refinement ratio of 2, 373,043,200 cells in total, 35 components, 98 GB on disk. The base mesh is $128 \times 128 \times 256$ with $\Delta x_0 = 14.53\,\mu$m. The unburnt temperature is $T_u = 693$ K and the unburnt fuel mass fraction is $Y_u = 1.325792 \times 10^{-2}$.

Two length scales govern what can be measured at which level. The thermal thickness, defined as $\delta_T = \Delta T / \max|\nabla T|$, is $19.4\,\mu$m at level 1 and $16.5\,\mu$m at level 2, giving

level	0	1	2	3
Δx (μm)	14.53	7.27	3.63	1.82
cells per δ_T	1.3	2.7	5.3	10.7

Quantities involving no derivatives — the lens thresholds, the enrichment, the explained variance — are converged at level 1. Curvature is a second derivative and is not: see Section 6.

2 Coarse-fine-aware gradients

Gradients are taken on the block-structured grid with the same treatment of coarse-fine interfaces that the solver uses, not on a derived or dual structure. A centred difference on a level's own ghost data is wrong at a coarse-fine interface in a specific way: the ghost cell is an interpolation of coarse data, so the stencil silently mixes two resolutions and the result is neither the coarse nor the fine gradient. The error sits exactly on the interface, which is a surface, so it does not converge away — it only gets thinner.

`Src/AMRGradient.cpp` instead builds a *composite* `MLPoisson` over the whole hierarchy, applies it, and takes the face-centred fluxes the operator computes, averaging them to cell centres. Two details are load-bearing:

- `mlmg.apply()` must precede `getFluxes`. The coarse-fine correction is established by the apply; `getFluxes` alone returns fluxes that have not seen it.
- `MLPoisson`'s flux is $-\nabla\phi$, not $\nabla\phi$.

This follows `Src/grad.cpp`. Note that `Src/curvature.cpp` builds its operators one level at a time inside a loop over levels, so its curvature, normals and strain carry the interface artefact described above — while, ironically, its optional progress-variable smoothing is a composite solve and therefore is coarse-fine aware.

2.1 Verification

`Tools/amrGradStatsTest.sh` runs 19 checks in each of 2D and 3D. The exact tests use a linear field, for which MLMG's interface interpolation is exact, so the only remaining error is floating point. Results are reported on three regions separately — all cells, interior cells, and the band within two cells of a coarse-fine interface — because a global norm cannot see a surface defect.

The gate is $\mathrm{tol} = 64\,\varepsilon\,|\phi|_{\max}/\Delta x$, and $/\Delta x^2$ for curvature, since a difference quotient divides by Δx. With the refined patch at the low corner the result is bit-exact; at the high corner the same code gives 7×10^{-15}, which is 32ε amplified by $1/\Delta x$ at $\Delta x = 1/128$.

Each test is paired with `per_level=1`, which builds one single-level operator per level — what `curvature.cpp` does. It must fail, and does, by a factor of 10^{14} relative to the tolerance, entirely within the coarse-fine band. A gate nothing can fail is not a gate.

3 The lens

3.1 Orientation

The lens is the region between crossing T and Y_{H_2} contours. In range space, T and Y_{H_2} are anti-correlated, so the data occupies a band running from (cold, rich) to (hot, lean). The quadrant $\{T > t_2\} \cap \{Y_{\mathrm{H}_2} < t_1\}$ is therefore the *burnt corner*: all products, no reaction. It gives an enrichment of $0.3\times$, worse than random.

The lens is the off-band quadrant,

$$L(t_1, t_2) = \{\, T > t_2 \,\} \cap \{\, Y_{\mathrm{H}_2} > t_1 \,\}, \tag{1}$$

gas that is simultaneously hotter *and* richer than the main band permits. This is the super-adiabatic crest.

3.2 Monotonicity

Writing $C(t_1, t_2)$ for the volume integral of the consumption density $q = -\dot{\omega}_{\mathrm{H}_2}$ over L, tightening either threshold shrinks the region. Measured on the 7641 occupied bins of the fine (T, Y_{H_2}) table, C is monotone in both arguments exactly, and *no bin has a negative conditional mean consumption density*, so the monotonicity is guaranteed rather than merely observed. The nested family $C_b < C_a$ is well posed.

3.3 Efficiency against the best possible region

A high R^2 for q given (T, Y_{H_2}) says a good function $\psi(T, Y_{\mathrm{H}_2})$ exists; it says nothing about whether ψ's level sets are shaped like quadrant corners. The sharp test is to compare the lens against

the *optimal* region of the same volume, which is a superlevel set of ψ and whose boundary is by definition a contour of consumption density.

budget	optimal	lens	$Y_{H_2} - Y_L(T)$	lens/opt	t_2 (K)	t_1	enrichment
0.20%	11.57%	10.23%	10.22%	0.884	1481	2.52×10^{-3}	51.4×
0.50%	25.68%	23.53%	22.54%	0.916	1469	1.83×10^{-3}	47.3×
0.87%	40.11%	37.27%	35.01%	**0.929**	1446	1.35×10^{-3}	42.9×
2.00%	69.32%	65.12%	60.69%	0.939	1268	8.63×10^{-4}	32.9×
5.00%	94.09%	93.03%	86.55%	0.989	889	3.11×10^{-4}	18.8×

The lens captures 93% of what the best possible region of identical volume captures, rising to essentially 100% at larger budgets. The pair of contours is very nearly the right shape, not merely a convenient one.

3.4 Optimality condition

For $L(t_1, t_2)$, maximising C at fixed volume V requires $\nabla C = \lambda \nabla V$. Since

$$\frac{\partial C}{\partial t_2} \Big/ \frac{\partial V}{\partial t_2} = \left. \frac{\int_{Y_{H_2} > t_1} q \, \mathrm{d}A}{\int_{Y_{H_2} > t_1} \mathrm{d}A} \right|_{T=t_2} = \langle q \rangle_{T\text{-face}}, \tag{2}$$

and likewise for the other face, the condition reduces to

$$\boxed{\langle q \rangle_{T\text{-face}} = \langle q \rangle_{Y\text{-face}} = \lambda} \tag{3}$$

The optimal corner sits where *both bounding faces lie on the same consumption-density level*, and λ is that level. This is the rigorous form of the observation that the contour intersection follows the consumption contours. Measured, the face-mean ratio is 1.00–1.12 over the useful range (median 1.07 for $\dot{\omega}_{H_2}$, 1.02 for heat release), departing only where t_1 reaches the lowest occupied bin.

3.5 A one-parameter family

There is no useful functional $t_1(t_2)$, because the optimum is nearly degenerate in t_2: a single fixed $t_2 \approx 1398$ K retains 0.917 of the full two-parameter optimum across budgets from 0.2% to 5%. A constant offset above the band fails — $t_1 - Y_L(t_2)$ scatters by 2.7 times its mean.

The better one-parameter form thresholds the excess fuel above the band, $Y_{H_2} - Y_L(T)$, where $Y_L(T) = \langle Y_{H_2} \mid T \rangle$. It matches the two-threshold lens and beats it at tight budgets. Every *linear* selector fails outright: T alone, Y_{H_2} alone, and the best rotated combination all capture under 1.3% of the consumption at the 0.87% budget, because the band is strongly curved. The structure is intrinsically a departure from the manifold, which is why a contour intersection works and no blended progress variable would.

3.6 What the angle of intersection does not add

The transversality measure $\sin\theta = |\nabla a \times \nabla b| / (|\nabla a||\nabla b|)$ vanishes wherever b is locally a function of a, for any monotone relation, and so is a pointwise, threshold-free measure of where a single progress variable fails. Adding it as a third coordinate raises R^2 for $\dot{\omega}_{H_2}$ from 0.867 to 0.873 — against a null control (`normalx_temp`, which by the x-periodicity of the domain should carry nothing) of +0.0002. The increment is real but negligible: given the *pair of values* (T, Y_{H_2}), the angle at which the contours cross says nothing further about the local consumption rate. The lens is characterised by which contours bound it, not by the geometry of the crossing.

4 The differential-diffusion postulate

4.1 Statement

With $\theta = (T - T_u)/(T_{ad} - T_u)$, $y = Y_{H_2}/Y_u$ and $D = \alpha/\mathrm{Le}$, subtracting the species and energy transport equations gives, for the excess-enthalpy scalar $S = \theta + y - 1$ (zero in the unburnt gas and in any $\mathrm{Le} = 1$ planar flame),

$$\frac{\partial S}{\partial t} + \mathbf{u} \cdot \nabla S = \nabla \cdot (\alpha \nabla S) + \nabla \cdot \big[\alpha(\tfrac{1}{\mathrm{Le}} - 1)\nabla y\big]. \tag{4}$$

The source is precisely the failure of the fuel diffusive flux and the conductive heat flux to cancel. Splitting the Laplacian into normal and surface parts, $\nabla^2 y = \partial_n^2 y + \kappa |\nabla y|$, the curvature contribution to the source is

$$\alpha \left(\frac{1}{\mathrm{Le}} - 1 \right) \kappa \, |\nabla y|. \tag{5}$$

Postulate. *The lens is the set $S > 0$: the region where differential diffusion has delivered more fuel than conduction has carried heat away. Its boundary is a level set of S, which to leading order is $Y_{H_2} - Y_L(T)$, and it forms at flame elements convex toward the reactants.*

With $\mathrm{Le} = 0.313$ the coefficient $(1/\mathrm{Le} - 1) = 2.23$ is of order unity: the effect is strong because hydrogen is this diffusive, not marginally so.

4.2 Test and result

The prediction is specific: at fixed temperature, $\langle Y_{H_2} \rangle$ must rise as κ becomes more negative. It holds at every temperature from 740 to 1878 K without exception, across 47 bins. At 1246 K,

curvature band	$\langle Y_{H_2} \rangle$	$\langle q \rangle$ ratio to cusps
$\kappa < -3 \times 10^4$ (crest)	5.47×10^{-3}	—
-3×10^4 to -1×10^4	5.00×10^{-3}	
-1×10^4 to 1×10^4	4.35×10^{-3}	
1×10^4 to 3×10^4	3.42×10^{-3}	
$\kappa > 3 \times 10^4$ (cusp)	2.46×10^{-3}	2.83

A factor of 2.2 in fuel mass fraction at identical temperature. Over the $\pm 10^5$ curvature range the induced spread is 5.6×10^{-3}, comparable to Y_L itself — an order-unity effect, not a perturbation. The consumption ratio between crests and cusps at fixed T is 2.6 through the preheat zone, rising to 4–8 through the reaction zone, so the chain $\kappa \to$ enrichment $\to$ consumption is visible at fixed temperature and is therefore not a consequence of crests merely being hotter.

4.3 The gradient scaling: supported, with a gap

If enrichment accumulates over a diffusive residence time $\tau \sim \delta^2/\alpha$, then integrating (5) gives $\mathrm{d}Y_{H_2}/\mathrm{d}\kappa \approx (1/\mathrm{Le} - 1)\,|\nabla Y_{H_2}|\,\delta^2$, so the ratio $(\mathrm{d}Y_{H_2}/\mathrm{d}\kappa)/\langle|\nabla Y_{H_2}|\rangle$ should be constant. Measured over 900–1600 K it is 9.57×10^{-11} m^2 with 31% scatter, while $|\nabla Y_{H_2}|$ itself varies by a factor of 2.3 — the constancy is the non-trivial part and it holds.

The implied length, however, is $\delta = 6.6\,\mu$m against a measured thermal thickness of $19.4\,\mu$m: a factor of 2.9 in δ, or 8.6 in δ^2. This is an order-of-magnitude consistency check that passed, not a quantitative confirmation. The residence-time step is dimensional analysis with an unspecified

constant, and the curvature is taken at level 1 where it is least reliable. Resolving this needs either trajectory integration or a level-3 repeat; see Section 10.

5 The Lewis number

5.1 Two estimators from the same first integrals

In the preheat zone, with constant mass flux $m = \rho u$, the species and energy equations integrate once to

$$\rho D \frac{\mathrm{d}Y_{\mathrm{H_2}}}{\mathrm{d}x} = m\,(Y_{\mathrm{H_2}} - Y_u), \qquad \lambda \frac{\mathrm{d}T}{\mathrm{d}x} = m\,c_p\,(T - T_u). \tag{6}$$

Writing $\xi = Y_u - Y_{\mathrm{H_2}}$ and $\tau = T - T_u$ and dividing,

$$\frac{\mathrm{d}\xi}{\mathrm{d}\tau} = \mathrm{Le}\,\frac{\xi}{\tau} \qquad \Longrightarrow \qquad \xi \propto \tau^{\mathrm{Le}}. \tag{7}$$

So the curvature of the T–$Y_{\mathrm{H_2}}$ band *is* a power law whose exponent is the Lewis number, and $\mathrm{Le} < 1$ is exactly what makes it concave. Two estimators follow, identical in a 1-D flame:

$$\mathrm{Le}^{(A)} = \frac{\mathrm{d}\ln\xi}{\mathrm{d}\ln\tau} \qquad \text{(band curvature)} \tag{8}$$

$$\mathrm{Le}^{(B)} = \frac{|\nabla Y_{\mathrm{H_2}}|}{|\nabla T|}\cdot\frac{\tau}{\xi} \qquad \text{(local flux ratio)} \tag{9}$$

5.2 They disagree, and why

estimator	preheat value	status
band curvature $\mathrm{Le}^{(A)}$	0.44	biased high by 45%
flux ratio, ξ from $\langle Y_{\mathrm{H_2}}\|T\rangle$	0.30	partly contaminated
flux ratio, ξ per bin	**0.313**	clean; IQR 0.281–0.342
literature, lean H_2–air	≈ 0.3	independent

The disagreement is not about Le. It is the statement

$$\frac{\mathrm{d}\langle Y_{\mathrm{H_2}}|T\rangle}{\mathrm{d}T} \;\neq\; \frac{\langle|\nabla Y_{\mathrm{H_2}}|\rangle}{\langle|\nabla T|\rangle}. \tag{10}$$

At $T \approx 770$ K the band slope is 2.19×10^{-5} and the mean-gradient ratio is 1.46×10^{-5}. The conditional mean $\langle Y_{\mathrm{H_2}}|T\rangle$ is not a path through a flame; it is an average over flame elements at different enrichment, so its slope is inflated by the very differential diffusion being measured. **The band-curvature estimator must not be used for** Le**.** It is retained in the tooling only because the *shape* it describes is what the lens exploits.

5.3 Apparent Lewis number as a diagnostic

Restricted to 713–1000 K and split by the sign of S:

region	volume	Le apparent
$S > 0$ (rich side, the lens)	1.060%	0.378
$S \leq 0$ (lean side, the cusps)	0.673%	0.279

This is not a transport coefficient varying — the mixture is the same. $\mathrm{Le}^{(B)}$ equals the true Lewis number only where the first integrals hold, and the postulate says they fail inside the lens by exactly the differential-diffusion source. Smaller ξ on the rich side inflates τ/ξ. The 35% split therefore *measures the imbalance* using only $\nabla Y_{\mathrm{H_2}}$, ∇T, T and $Y_{\mathrm{H_2}}$, with no reference to reaction rates, curvature or heat release. The true transport value lies between, near 0.31.

5.4 Temperature dependence

$\mathrm{Le}^{(B)}$ rises from 0.31 at 723 K to 0.43 at 1374 K. This is *not* the temperature dependence of the transport coefficients: for $\mathrm{H_2}$ in $\mathrm{N_2}$, $D \propto T^{1.7}$ and $\alpha = \lambda/\rho c_p \propto T^{1.7}/c_p$, so $\mathrm{Le} \propto 1/c_p$ should fall slightly. The observed rise tracks heat release, as the no-reaction assumption behind the first integrals fails.

6 Curvature and resolution

Curvature is computed as $\kappa = \nabla\cdot\mathbf{n}$ with the normal regularised as

$$\mathbf{n} = \frac{\nabla T}{\max\big(|\nabla T|,\ f\big)}, \qquad f = 10^{-2}\max|\nabla T|, \tag{11}$$

dividing by $\max(|\nabla T|, f)$ rather than zeroing below a floor. Hard zeroing makes the normal field jump from zero to unit length across the floor contour, and the divergence of that jump is a spurious curvature sitting on a level set of $|\nabla T|$ — an artefact that looks exactly like a flame structure. The regularised form is continuous everywhere.

The floor governs 66.27% of the volume but only 0.16% of the consumption, so it costs nothing.

The extremum is mesh noise at every level.

	level 1	level 2	ratio
Δx	$7.27\,\mu$m	$3.63\,\mu$m	0.50
$\max\lvert\nabla T\rvert$	7.78×10^{7}	9.12×10^{7}	1.17
$\max\lvert\kappa\rvert$	3.96×10^{5}	7.78×10^{5}	**1.96**
$\max\lvert\kappa\rvert\cdot\Delta x$	**2.88**	**2.83**	—

Halving Δx doubled $\max|\kappa|$. The first derivative is converging (it grew by 1.17×, not 2×); the second derivative's extremum is not, and will keep doubling. This is expected of any finite-difference second derivative and means **the maximum is the wrong statistic**: use the volume-weighted distribution, read where there is a flame. At level 2, 63% of the flame volume has $|\kappa| < 3\times10^{4}$, a radius above $33\,\mu$m or nine cells, which is adequately resolved.

On smoothing. `curvature.cpp` offers a pre-smoothing step, $(I-\beta\nabla^2)\tilde{\phi} = \phi$, solved compositely. It would produce finite, smooth, plausible curvatures — which is the danger, since their values would be set by β rather than by the flame. At the default `smoothing_time` $= 10^{-7}$, $\sqrt{\beta} = 316\,\mu$m against a $19.4\,\mu$m flame in a 1.86 mm box: it would erase the cellular pattern. Matching the flame scale needs $\beta \approx \delta_T^2 \approx 3.8\times10^{-10}$, some 250× smaller. More fundamentally, the high-$|\kappa|$ tail *is* the thermodiffusive signature, so smoothing attenuates exactly what is being measured. It is the right tool for the cell envelope at a chosen scale; it is the wrong tool for recovering curvature the mesh did not resolve. If used, sweep β and report what is stable.

Sign convention. `amrGradStats` uses **n** pointing toward the *products*; `curvature.cpp` and the combustion literature point it toward the reactants. The two curvatures are therefore negatives of one another. In this document $\kappa < 0$ is a crest, convex toward the reactants. In the standard convention the central result reads: positive curvature burns roughly ten times faster than negative.

7 Synthesis

The three threads are one mechanism.

- The **band curvature** in the T–Y_{H_2} plane is the power law $\xi \propto \tau^{\mathrm{Le}}$, so $\mathrm{Le} = 0.313$ sets its shape.
- The **lens** is a level set of $S = \theta + y - 1$, which is the accumulated differential-diffusion imbalance. This is why the one-parameter selector $Y_{H_2} - Y_L(T)$ matches a two-threshold family, and why every linear selector fails.
- The **curvature–consumption correlation** is the source of S, with coefficient $(1/\mathrm{Le} - 1) = 2.23$.

The 47× consumption enrichment found by sweeping thresholds is that mechanism seen through a quadrant approximation.

This also answers the question the effort began with. A cell is a connected component of $\{S > s\}$: one threshold rather than two, monotone by construction, and — once $Y_L^0(T)$ comes from a 1-D flame rather than from the field itself — meaning the same thing in every snapshot, which is the property needed to relate elements through time.

8 Procedure

8.1 Build

```
cd $PELE_ANALYSIS/Src
for t in amrComplexVerify amrContourVerify amrSweep amrJointFit \
        makeTestPlt surfMEFtoDAT amrGradStats amrGradVerify ; do
    make -f GNUmakefile.AMRComplex AMREX_HOME=$AMREX_HOME \
        USE_MPI=TRUE COMP=gnu DIM=3 EBASE=$t -j16
done
```

`amrGradStats` and `amrGradVerify` are the only targets that pull in `Boundary`, `AmrCore` and `LinearSolvers`; the makefile adds those directories only for them. The first such build spends several minutes compiling AMReX.

A recurring failure. A driver must not be the case-variant of a library source. On a case-insensitive filesystem `amrComplex.cpp` *is* `AMRComplex.cpp`: the second file written replaces the first, one object is built instead of two, and the link fails on every library symbol with "referenced from _main". This has happened twice in this tree. Hence `amrComplexVerify.cpp` and `amrJointFit.cpp`, and hence the name guard in `Tools/amrBuildAndTest.sh`, which checks before anything is compiled. A comment is not a check.

8.2 Verify

```
cd $PELE_ANALYSIS
Tools/amrBuildAndTest.sh
DIM=3 Tools/amrGradStatsTest.sh
DIM=2 Tools/amrGradStatsTest.sh
```

Expect 19 passes in each dimension from the gradient suite. Do not proceed on a failure: each check is paired with a control that must fail, so a suite that passes vacuously would be silent.

8.3 Run

```
Tools/runLensPipeline.sh --infile /path/to/plt29791 --case lean-h2 \
    --tu 693 --yu 1.325792e-2 --finest 1 --isper "1 1 0" --nproc 16
```

T_u and Y_u are case properties and must be the prescribed inflow values — the Lewis estimates are very sensitive to them ($\mathrm{Le}^{(A)}$ moves from 0.44 to 0.81 if T_u is taken as 680 rather than 693 K).

The pipeline writes, into `lens_<case>_L<n>/`: the derived plotfile; four `amrJointFit` tables; and the analysis text files `frontier.txt`, `band.txt`, `lewis.txt`, `postulate.txt` and `r2_gated.txt`. Stages may be run separately with `--stage grad|stats|analyse`.

8.4 Analyse an existing table

```
Tools/lensAnalysis.py frontier ty_fine.dat --resp "I_R(H2)"
Tools/lensAnalysis.py lewis lewis2d.dat --Tu 693 --Yu 1.325792e-2
Tools/lensAnalysis.py postulate postulate.dat
Tools/lensAnalysis.py r2 ty_curv_gated.dat
```

Columns are read from the header line rather than by position, so each mode works on any `amrJointFit` table that carries the variables it needs.

9 Porting to a larger machine

9.1 Resources

Derived from the 373M-cell case: approximately 2.8 GB of output per component at full depth, and peak resident memory of about 20 component-equivalents (the state array plus per-variable scratch, face MultiFabs and the Laplacian).

`finestLevel`	cells	stage-1 output	stage-1 peak RSS
1	25 M	∼4 GB	∼6 GB
2	121 M	∼17 GB	∼30 GB
3	373 M	∼53 GB	∼95 GB

Component count is the lever: dropping a variable from `vars` to `carry` removes its gradient, magnitude and all its alignment pairs.

9.2 GPU status, honestly

component	state	note
`AMRGradient`	GPU-ready	MLMG apply, `getFluxes` and `average_face_to_cellcenter` are all device-resident
`amrGradStats`	GPU-ready	pointwise work is `ParallelFor`; one host loop remains, see below
`AMRJointStats`	host only	`forEachNode` with a host lambda and Kahan compensation into a host array
`AMRComplex`	host only	nine host loops in the traversal

This split is fortunate: the expensive step at scale is the composite MLMG apply, which is the GPU-capable part.

Known blocker. The normal-floor volume diagnostic in `amrGradStats.cpp` uses `AMREX_LOOP_3D`, a host loop, over data that under a CUDA or HIP build lives in device memory. It is a reduction and should become a `ReduceSum`. It has not been converted, because it has not been possible to test a GPU build here. **Before trusting any GPU result, build with `USE_CUDA=TRUE` and run `Tools/amrGradStatsTest.sh`**; if that diagnostic faults or reports a nonsensical volume fraction, this is why, and the remedy is either the `ReduceSum` conversion or guarding the diagnostic out under `AMREX_USE_GPU`.

GPU build.

```
make -f GNUmakefile.AMRComplex AMREX_HOME=$AMREX_HOME \
     USE_MPI=TRUE USE_CUDA=TRUE COMP=gnu DIM=3 EBASE=amrGradStats -j16
```

One rank per GPU. The histogram stage of `AMRJointStats` would port as a `ParallelFor` plus `Gpu::Atomic::Add` into a table of order 1 MB, but that work has not been done.

9.3 Transfer checklist

1. Clone AMReX at a known commit; this work was developed against `65db2e4e6`.
2. Clone PeleAnalysis, branch `amr-complex`.
3. Build the eight targets listed above; confirm the name guard passes.
4. Run `Tools/amrBuildAndTest.sh` and both dimensions of `Tools/amrGradStatsTest.sh`. All must pass before any science.
5. Copy or regenerate the plotfile. Record T_u and Y_u from the inputs file, not from the data.
6. Run the pipeline at `--finest 1` first. Compare `frontier.txt` and `lewis.txt` against Sections 3 and 5. They should agree to the digits quoted.
7. Only then go to `--finest 3`.

10 Open items

1. $Y_L^0(T)$ **from a 1-D laminar flame.** This is the weakest link. $Y_L(T)$ currently comes from the field under analysis, so it defines S, sets the lens threshold, and biased $\mathrm{Le}^{(A)}$ by 45% — all while containing the signal being measured. A Cantera or 1-D PeleLMeX solution at the case conditions removes the self-reference, makes the lens threshold portable across snapshots, and gives $\mathrm{Le}(T)$ from transport data as a fourth independent check on 0.313.

2. **Curvature at level 3.** Everything in Section 4 uses level 1, where $\max|\kappa| = 2.85/\Delta x$. Noise degrades a correlation rather than creating one, so the monotone trend is safe, but the slope -2.98×10^{-8} — and hence the 6.6 μm that should have been 19.4 μm — will move with resolution.

3. **Residence time.** The δ^2 in Section 4 is dimensional analysis. Integrating (5) along trajectories would turn an order-of-magnitude check into a quantitative one.

4. **Time sequence.** Everything here is one snapshot. The purpose of a reference-free, single-threshold cell definition is to track elements between snapshots; that has not been attempted.

5. **GPU.** See Section 9.